ASSIGNMENT NO. 2-Study of XML and XSD

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Problem Statement 1

1. What is XML? Why is it called an Extensible Markup Language?

XML (Extensible Markup Language) is a markup language used to store and transport data in a structured and readable format. It is called *extensible* because users can define their own custom tags according to application needs, unlike HTML which has predefined tags.

2. Difference between Well-Formed and Valid XML + Structure

Well-Formed XML

- Follows XML syntax rules
- Proper opening and closing tags
- Correct nesting

Valid XML

- Well-formed and Follows rules defined in DTD or XSD.

Basic XML Structure Example

```
<?xml version="1.0" encoding="UTF-8"?>
<student>
    <name>Jatin</name>
    <branch>CSE</branch>
</student>
```

3. Difference between XML and HTML

XML	HTML
Used to store data	Used to display data
User-defined tags	Predefined tags

Strict syntax	Lenient syntax
Platform independent	Browser dependent

XML is preferred for data storage because it is structured, flexible, and platform independent.

4. What is XSD? Why preferred over DTD?

XSD (XML Schema Definition) defines structure, data types, and constraints of an XML document.

Why XSD over DTD

- Supports data types
- Written in XML
- Better validation
- Namespace support

Roles

- xsd:schema → Root of schema
- xsd:element → Defines elements
- xsd:complexType → Defines nested elements

5. Difference between simpleType and complexType

simpleType	**complexType**
No child elements	Can contain child elements
Used for values	Used for structures

Example

```
<xsd:simpleType name="yearType">
  <xsd:restriction base="xsd:string">
    <xsd:enumeration value="Second"/>
    <xsd:enumeration value="Third"/>
```

```
    <xsd:enumeration value="Fourth"/>
  </xsd:restriction>
</xsd:simpleType>
```

6. Linking XML with XSD & Role of xsi

XML is linked to XSD using:

xsi:noNamespaceSchemaLocation="student.xsd"

xsi (XML Schema Instance) tells the parser where the schema is located.

7. How XSD enforces data constraints

Example 1: Restrict phone length

```
<xsd:restriction base="xsd:string">
  <xsd:length value="10"/>
</xsd:restriction>
```

Example 2: Allowed year values

```
<xsd:enumeration value="Second"/>
```

8. minOccurs and maxOccurs

- minOccurs → Minimum occurrences
- maxOccurs → Maximum occurrences

They control how many times an element can appear during validation.

9. XML well-formed but not valid – Reasons

- Missing XSD
- Data type mismatch
- Invalid value (not allowed by schema)
- Missing mandatory element

10. Real-world usage & XML vs JSON

Uses

- Web services (SOAP)
- Configuration files
- Banking systems
- Data exchange

XML	JSON
Verbose	Lightweight
Schema based	Faster
Used in enterprise apps	Used in APIs

Problem Statement 2

Task 1: student.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<students xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:noNamespaceSchemaLocation="student.xsd">

   <student>
      <studentId>101</studentId>
      <name>Jatin</name>
      <year>Second</year>
      <branch>CSE</branch>
      <email>jatin@gmail.com</email>
      <phone>9876543210</phone>
   </student>

   <student>
      <studentId>102</studentId>
      <name>Rahul Patil</name>
      <year>Third</year>
      <branch>CSE</branch>
      <email>rahul@gmail.com</email>
      <phone>9123456789</phone>
   </student>

</students>
```

Task 2: student.xsd

```
<?xml version="1.0" encoding="UTF-8"?>
<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema">

   <xsd:element name="students">
      <xsd:complexType>
         <xsd:sequence>
            <xsd:element name="student" maxOccurs="unbounded">
               <xsd:complexType>
                  <xsd:sequence>

                     <xsd:element name="studentId" type="xsd:integer"/>

                     <xsd:element name="name" type="xsd:string"/>

                     <xsd:element name="year">
                        <xsd:simpleType>
                           <xsd:restriction base="xsd:string">
                              <xsd:enumeration value="Second"/>
                              <xsd:enumeration value="Third"/>
                              <xsd:enumeration value="Fourth"/>
                           </xsd:restriction>
                        </xsd:simpleType>
                     </xsd:element>

                     <xsd:element name="branch" type="xsd:string"/>

                     <xsd:element name="email" type="xsd:string"/>

                     <xsd:element name="phone">
                        <xsd:simpleType>
                           <xsd:restriction base="xsd:string">
                              <xsd:length value="10"/>
                           </xsd:restriction>
                        </xsd:simpleType>
                     </xsd:element>

                  </xsd:sequence>
               </xsd:complexType>
            </xsd:element>
         </xsd:sequence>
      </xsd:complexType>
   </xsd:element>

</xsd:schema>
```

Task 3: Linking XML with XSD

Linked using:

xsi:noNamespaceSchemaLocation="student.xsd"

Task 4: Validation

The XML file was validated successfully using **VS Code XML extension**, and no errors were found.

Outputs:

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<students xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="student.xsd">
  ▼<student>
      <studentId>101</studentId>
      <name>Jatin</name>
      <year>Second</year>
      <branch>CSE</branch>
      <email>jatin@gmail.com</email>
      <phone>9876543210</phone>
    </student>
  ▼<student>
      <studentId>102</studentId>
      <name>Rahul Patil</name>
      <year>Third</year>
      <branch>CSE</branch>
      <email>rahul@gmail.com</email>
      <phone>9123456789</phone>
    </student>
  </students>
```